SA1 SECTION I

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2026-10-11

I.

1. A company has three factories producing a product. Factory 1 produces x1 of the product, factory 2 produces x2, and factory 3 produces x3, where the sum of these is 1. The defective rates of the products are y1, y2, and y3 respectively, where the sum of these is 0.12. Write a program (user input for xi and yi) to calculate the probability that a randomly selected product is defective.

```
> getval <- function(prompt, n = 3) {
+   repeat {
+     vals <- scan(
+       what = numeric(),
+       nmax = n,
+       quiet = TRUE
+     )
+
+     if (lenght(vals) == n && all(is.finite(vals))) {
+       return(vals)
+     }
+
+     cat(prompt, "\n")
+     cat("Enter", n,"valid numbers separated by spaces. \n")
+   }
+ }
> repeat {
+   cat("Enter x1, x2, and x3 (0.10 to 0.40 each):\n")
+   x <- scan(what = numeric(), nmax = 3, quiet = TRUE)
+
+   if (length(x) == 3 &&
+       all(is.finite(x)) &&
+       all(x >= 0.10 & x <= 0.40) &&
+       abs(sum(x) - 1) < 1e-8) {
+     break
+   }
+
+   cat("Invalid input. 0.10 < xi < 0.40, and the sum must equal to 1.\n")
+ }
Enter x1, x2, and x3 (0.10 to 0.40 each):
1: 0.30 0.35 0.35
```

Figure 1: Input x1, x2, and x3

```
> repeat {
+   cat("Enter y1, y2, and y3 (0.01 to 0.05 each):\n")
+   y <- scan(what = numeric(), nmax = 3, quiet = TRUE)
+
+   if (length(y) == 3 &&
+       all(is.finite(y)) &&
+       all(y >= 0.01 & y <= 0.05) &&
+       abs(sum(y) - 0.12) < 1e-8) {
+     break
+   }
+
+   cat("Invalid input. 0.01 < yi < 0.05, and the sum must equal to 0.12.\n")
+ }
Enter y1, y2, and y3 (0.01 to 0.05 each):
1: 0.0356 0.0409 0.0435
> # Probability of a Defective Product
>
> prb <- sum(x * y)
>
> cat("\nProductions:", x, "\n")

Productions: 0.3 0.35 0.35
>
> cat("\nDefective Rates:", y, "\n")

Defective Rates: 0.0356 0.0409 0.0435
>
> cat("\nProbability of a Defective Product Selected:", prb, "or", prb*100, "%\n")

Probability of a Defective Product Selected: 0.04022 or 4.022 %
```

Figure 2: Input y1, y2, and y3

The user input 0.3, 0.35, and 0.35 for the production of the 3 factories. All of these ranged between 0.1 and 0.4 and equals to 1 when added. Their defective rates are: 0.0356, 0.0409, and 0.0435, respectively. These rates are in between of 0.01 and 0.05 and totalled to 0.12. The probability of picking a defective product is 0.04022 or 4.022%.

2. With your own computing experience, develop a front end to R that allows the user to input the values of a univariate discrete random variable and the associated probabilities and to obtain the mean and variance, and to input the values of a bivariate discrete random variable and the associated probabilities and to obtain the marginal and conditional distributions. Your program should provide a facility to calculate the mean and variance of each distribution, and to plot the pdf and cdf. In each program, do validity checks that the probabilities are in the interval [0, 1], and that they sum to one.

UNIVARIATE DISTRIBUTION

```
 DISCRETE RANDOM VARIABLES

1. Univariate Distribution
2. Bivariate Distribution
3. Exit
Choose an option: 1
Number of values: 3
Enter 3 values (separated by spaces):
1: 1 2 3
Enter 3 probabilities (separated by spaces):
1: 0.3 0.35 0.35

Univariate Distribution:
  Value Probability
1     1        0.30
2     2        0.35
3     3        0.35

Mean and Variance:
    Mean Variance
  2.0500   0.6475

 DISCRETE RANDOM VARIABLES

1. Univariate Distribution
2. Bivariate Distribution
3. Exit
Choose an option: 3
Program ended.
```

Figure 3: Univariate Distribution

Reusing the percentage of each selecting products from each factory from the previous item, the graphs show their probability, with the two 0.35 having a higher chance compared to the 0.30. Meanwhile, the other graph depicts the totality that adds up from the lowest value (0.3) until the maximium value.

BIVARIATE DISTRIBUTION

The user intended to try the probabilities of tossing a coin 2x, with X represents the 1st toss and the Y is the 2nd toss, and 0 and 1 as heads and tails, respectively. Marginal distribution shows that all are possible outcomes and has the probability of 0.25. Both 1st and 2nd toss have 0.5 chance of getting a head or a tail,

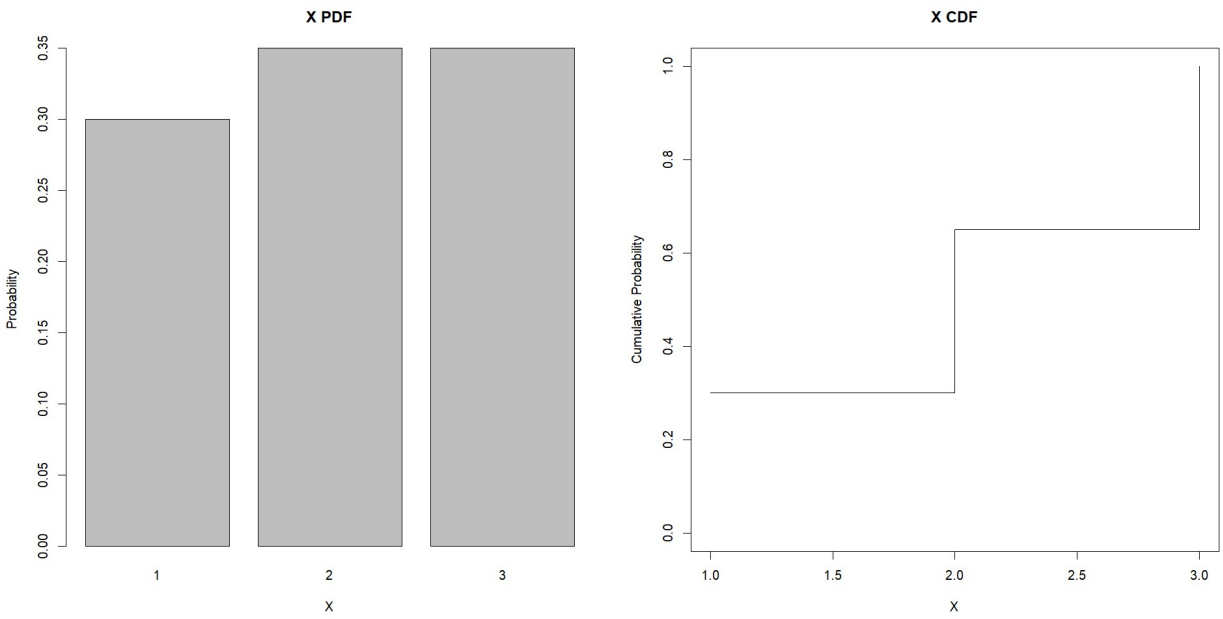


Figure 4: Univariate Distribution - PDF & CDF

and both have the same mean (0.5) and variance (0.25). The conditional distribution tells us that the 1st toss does not affect the 2nd toss and vice versa.

3. By generating 10,000 searches in R, carry out a simulation experiment for a search engine going through a list of sites for a given key phrase, until the key phrase is found. You should allow your program to input the probability p that any site will contain the key phrase.

Plot the simulated pdf and calculate its mean and variance, and obtain the simulated conditional distribution of searches when three searches have been carried out without success. Calculate its mean and variance, and satisfy yourself that they are equivalent to the simulated distribution of the complete set. As test data assume each site has a 60% chance of containing the key phrase. To satisfy yourself that the Markov memoryless property holds, obtain estimates of (a) P(X=4|X>3) and P(X=1), and (b) P(X=5|X>3) and P(X=2) where X is the number of searches to the first success.

With a success chance of 60%, is the same as the theoretical probability of the first search being successful, while on the second search is 24%. The graph shows that the probability of the first success decreases as the required number of searches increases. The expected number of searches is about 1.67 and the variance is around 1.11, which are close to the simulated results. By the Memoryless Property, the probability of success on the next try is still 0.6, disregarding the previous failed attempts, which is the same for the next search at 0.24. This suggests that the theoretical probabilities before and after 3 failures equal to the corresponding waiting times. But the simulated ones are not exactly equal as this consists of finite random trials.

```
 DISCRETE RANDOM VARIABLES

1. Univariate Distribution
2. Bivariate Distribution
3. Exit
Choose an option: 2
Number of X values: 2
Number of Y values: 2
Enter 2 X values (separated by spaces):
1: 0 1
Enter 2 Y values (separated by spaces):
1: 0 1

Enter the Joint Probability Matrix row by row.
Rows correspond to X; columns correspond to Y.
Probabilities for X = 0 (enter 2 numbers):
1: 0.25 0.25
Probabilities for X = 1 (enter 2 numbers):
1: 0.25 0.25

Joint Probability Distribution:
     Y=0  Y=1
X=0 0.25 0.25
X=1 0.25 0.25

Marginal Distribution of X:
    X Probability
X=0 0         0.5
X=1 1         0.5

Marginal Distribution of Y:
    Y Probability
Y=0 0         0.5
Y=1 1         0.5
```

Figure 5: Bivariate Distribution

```
Mean and Variance of X:
    Mean Variance
    0.50     0.25

Mean and Variance of Y:
    Mean Variance
    0.50     0.25

P(Y | X = 0 )
    Value Probability
Y=0     0         0.5
Y=1     1         0.5
    Mean Variance
    0.50     0.25

P(Y | X = 1 )
    Value Probability
Y=0     0         0.5
Y=1     1         0.5
    Mean Variance
    0.50     0.25

P(X | Y = 0 )
    Value Probability
X=0     0         0.5
X=1     1         0.5
    Mean Variance
    0.50     0.25

P(X | Y = 1 )
    Value Probability
X=0     0         0.5
X=1     1         0.5
    Mean Variance
    0.50     0.25
```

Figure 6: Bivariate Distribution

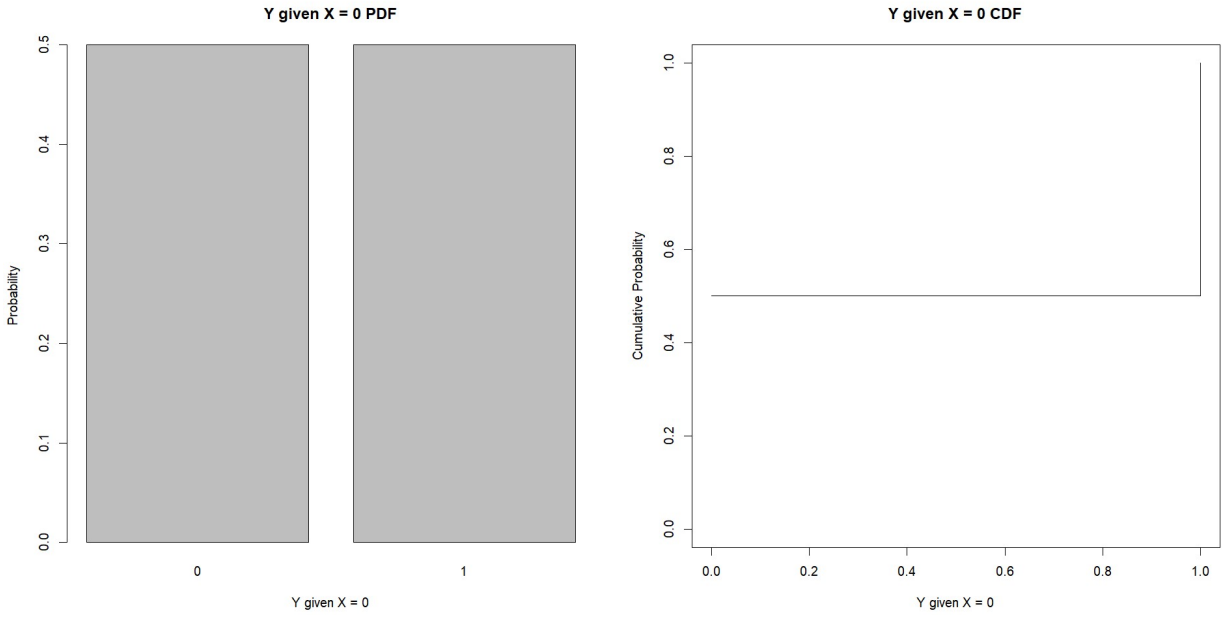


Figure 7: Bivariate Distribution - X0

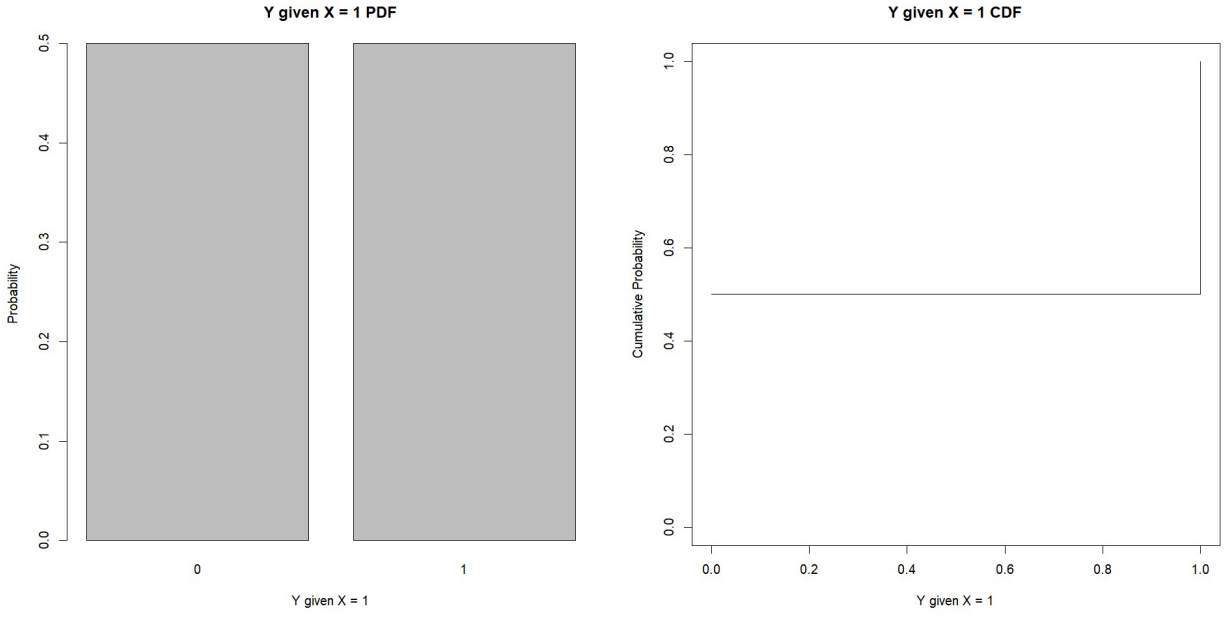


Figure 8: Bivariate Distribution - X1

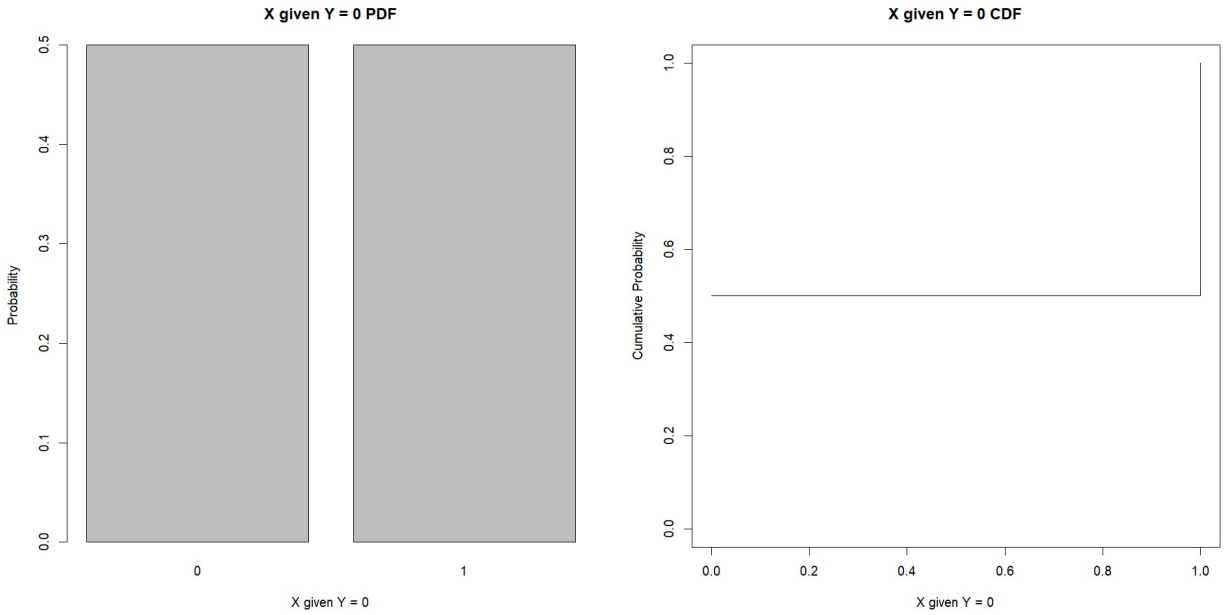


Figure 9: Bivariate Distribution - Y0

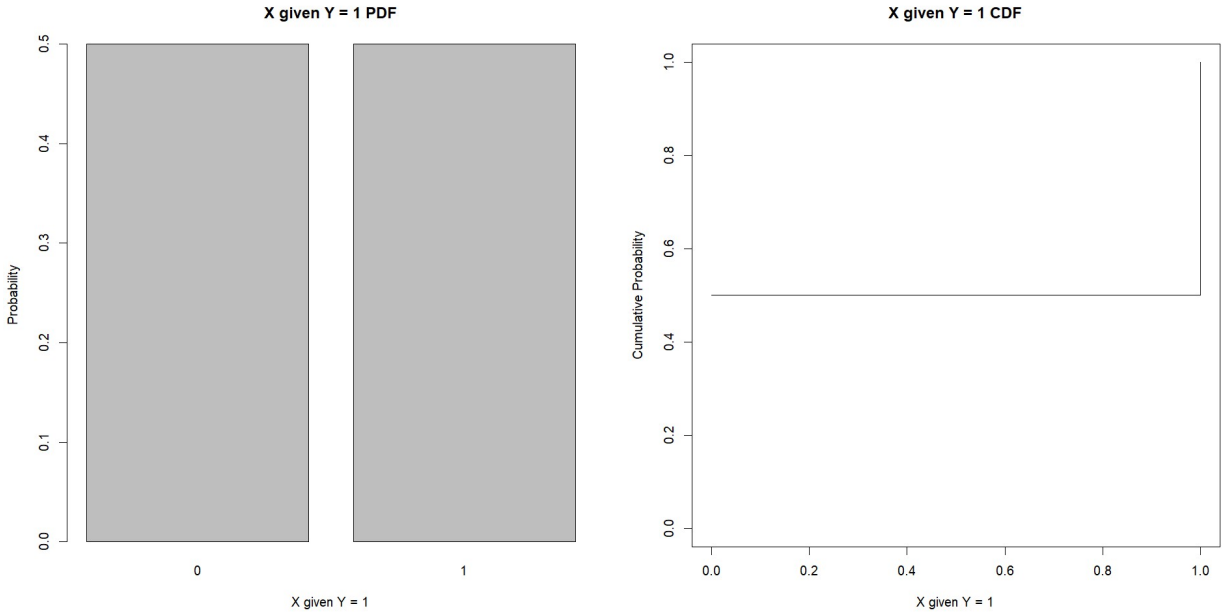


Figure 10: Bivariate Distribution - Y1

```
Probability of Success p (0 < p <= 1): 0.6
> X <- rgeom(n, prob = p) + 1
>
> pdftable <- prop.table(table(X))
> cat("Simulated PDF\n")
Simulated PDF
> pdftable
X
     1      2      3      4      5      6      7      8      9     11
0.6064 0.2352 0.0932 0.0378 0.0173 0.0063 0.0024 0.0008 0.0004 0.0002
>
> cat("Mean:", mean(X), "\n")
Mean: 1.6609
> cat("Variance:", var(X), "\n")
Variance: 1.117023
>
> # Theoretical
>
> cat("Mean:", 1/p, "\n")
Mean: 1.666667
> cat("Variance:", (1 - p) / p^2, "\n")
Variance: 1.111111
```

Figure 11: Simulated PDF

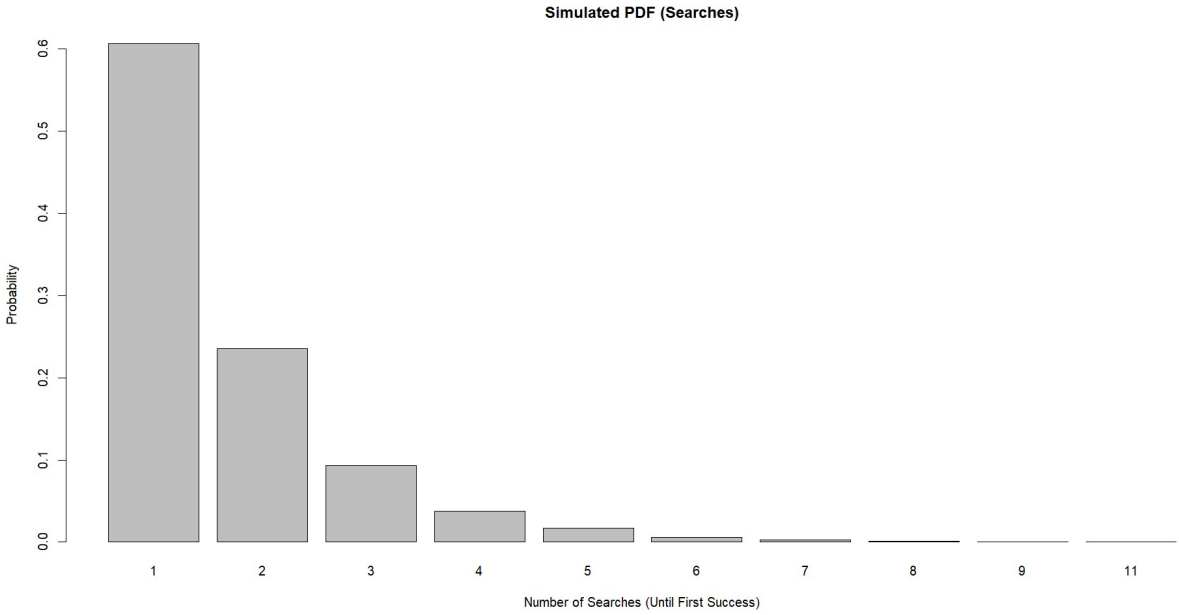


Figure 12: Simulated PDF - Graph

```
> X_cond <- X[X > 3]
> Y <- X_cond - 3
> cat("\nNumber of Simulations (X>3):", length(X_cond), "\n")

Number of Simulations (X>3): 652
>
> cat("\nConditional Distribution of Remaining Searches:\n")

Conditional Distribution of Remaining Searches:
> cond <- prop.table(table(Y))
> print(cond)
Y
          1           2           3           4           5           6           8
0.579754601 0.265337423 0.096625767 0.036809816 0.012269939 0.006134969 0.003067485
>
> cat("Mean:", mean(Y), "\n")
Mean: 1.670245
> cat("Variance:", var(Y), "\n")
Variance: 1.035488
>
> cat("Mean and Conditional Mean: ", mean(X), "\t", mean(Y), "\n")
Mean and Conditional Mean:  1.6609       1.670245
> cat("Variance and Conditional Variance: ", var(X), "\t", var(Y), "\n")
Variance and Conditional Variance:  1.117023     1.035488
>
```

Figure 13: Conditional PDF

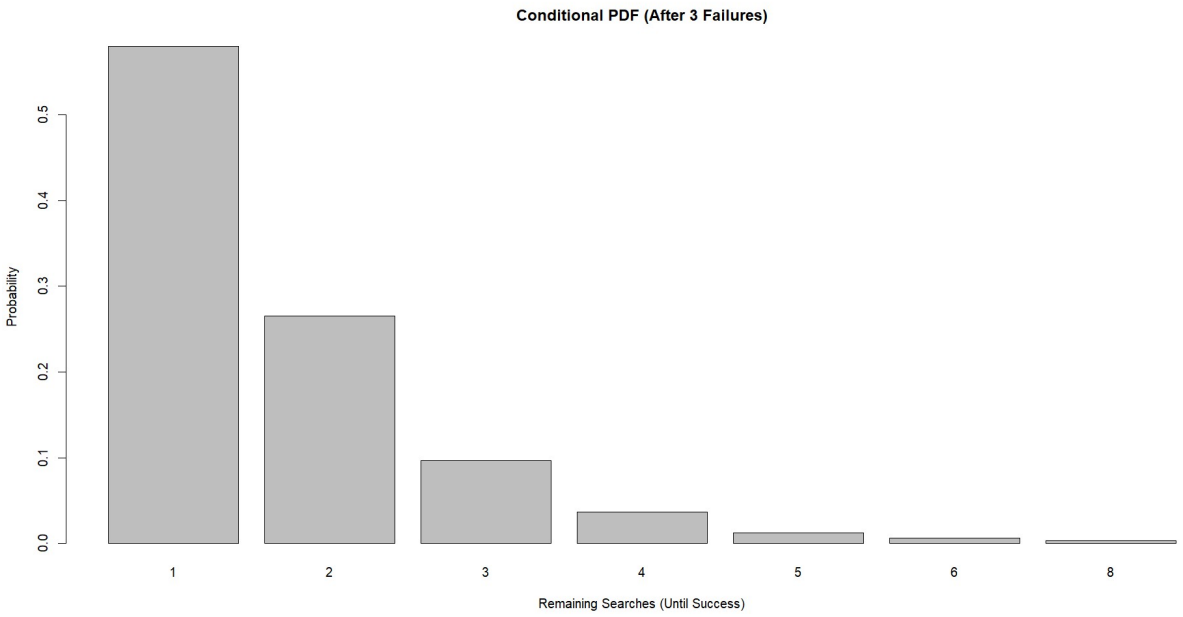


Figure 14: Conditional PDF - Graph

```
> # Markov Memoryless Property
>
> # (a) P(X=4|X>3) and P(X=1)
>
> p4g3 <- sum(X == 4) / sum(X > 3)
> prb1 <- mean(X == 1)
>
> # (b) P(X=5|X>3) and P(X=2)
>
> p5g3 <- sum(X == 5) / sum(X > 3)
> prb2 <- mean(X == 2)
>
> cat("\nP(X = 4 | X > 3):", p4g3, "\n")

P(X = 4 | X > 3): 0.5797546
> cat("P(X = 1):", prb1, "\n")
P(X = 1): 0.6064
>
> cat("\nP(X = 5 | X > 3):", p5g3, "\n")

P(X = 5 | X > 3): 0.2653374
> cat("P(X = 2):", prb2, "\n")
P(X = 2): 0.2352
>
> # Theoretical Probabilities
>
> cat("P(X = 4 | X > 3) = p(1-p)^3 / (1-p)^3 =", p, "\n")
P(X = 4 | X > 3) = p(1-p)^3 / (1-p)^3 = 0.6
> cat("P(X = 1) =", p, "\n")
P(X = 1) = 0.6
>
> cat("P(X = 5 | X > 3) = p(1-p)^4 / (1-p)^3 =", p * (1-p), "\n")
P(X = 5 | X > 3) = p(1-p)^4 / (1-p)^3 = 0.24
>
> cat("P(X = 2) =", p * (1-p), "\n")
P(X = 2) = 0.24
```

Figure 15: Markov Memoryless Property